MH1 1541A H15 Test 3 V1 Solutions

1. If the coefficient matrix A in a homogeneous system of 16 equations in 20 unknowns is known to have rank 10, how many parameters are there in the general solution?

A. none
B. 4
C. 6
(D.) 10
E. 16
F. 20

(Homogeneous ∴ consistent).
So # parameters = # variables − rank A
= $20 - 10$
= 10

2. If $A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}$, and B is a $3 \times n$ matrix then the third row of the matrix AB is

A. the same as the second row of A.
B. the same as the first row of B.
C. the same as the second row of B.
D. the sum of the first and the second rows of B.
E. the sum of the first and the third rows of B.
(F.) the sum of the second and third rows of B.

Write $B = \begin{bmatrix} r_1 \\ r_2 \\ r_3 \end{bmatrix}$ in block row form (r_i = i^{th} row of B)

Then $AB = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} r_1 \\ r_2 \\ r_3 \end{bmatrix} = \begin{bmatrix} r_1 \\ r_1 + r_2 + r_3 \\ r_2 + r_3 \end{bmatrix}$

b C_1 C_2 C_3

3. Find the value(s) of t for which $(1, 2, 3, t)$ lies in the subspace spanned by $(1, 0, 1, 2)$, $(0, 1, 1, 2)$ and $(1, 1, 0, 2)$.

A. $t = 4$ or 6
B. $t = 4$ only
C. $t = 6$ only
D. $t = -2$ or -4
E. $t = 0$ or 2
F. $t = -2$, 0 or 4

$[C_1\ C_2\ C_3 \mid b]$ consistent $\Leftrightarrow$ $b \in \text{span}\{C_1, C_2, C_3\}$:

$$\left[\begin{array}{ccc|c} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 2 \\ 1 & 1 & 0 & 3 \\ 2 & 2 & 2 & t \end{array}\right] \sim \left[\begin{array}{ccc|c} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 2 \\ 0 & 1 & -1 & 2 \\ 0 & 2 & 0 & t-2 \end{array}\right]$$

$$\sim \left[\begin{array}{ccc|c} 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & t/2 - 1 \\ 0 & 0 & -1 & -t/2 + 3 \\ 0 & 0 & 1 & -t/2 + 3 \end{array}\right] \sim \left[\begin{array}{ccc|c} 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & t/2 - 1 \\ 0 & 0 & -1 & -t/2 + 3 \\ 0 & 0 & 0 & -t + 6 \end{array}\right]$$

This is consistent $\Leftrightarrow$ $t = 6$

4. Suppose $e, f \in \mathbf{R}$ and consider the linear system in x, y and z:

$$\begin{array}{rcrcrcl} 2x & - & 2y & + & ez & = & f \\ x & & & + & z & = & -1 \\ 3x & + & y & + & 2z & = & -1 \end{array}$$

a) If $[A\,|\,b]$ is the augmented matrix of the system above, find $\operatorname{rank} A$ and $\operatorname{rank}[A\,|\,b]$ for all values of e and f.

$$[A\,|\,b] = \left[\begin{array}{ccc|c} 2 & -2 & e & f \\ 1 & 0 & 1 & -1 \\ 3 & 1 & 2 & -1 \end{array}\right] \sim \left[\begin{array}{ccc|c} 1 & 0 & 1 & -1 \\ 0 & 1 & -1 & 2 \\ 0 & -2 & e-2 & f+2 \end{array}\right]$$

$$\sim \left[\begin{array}{ccc|c} 1 & 0 & 1 & -1 \\ 0 & 1 & -1 & 2 \\ 0 & 0 & e-4 & f+6 \end{array}\right].$$

Hence $\operatorname{rank} A = \begin{cases} 2 & \text{if } e = 4 \\ 3 & \text{if } e \neq 4 \end{cases}$

$\operatorname{rank}[A\,|\,b] = \begin{cases} 2 & \text{if } e = 4 \text{ and } f = -6 \\ 3 & \text{otherwise.} \\ & (e \neq 4 \text{ or } f \neq -6) \end{cases}$

(Q.4 parts (b) and (c) are on the next page...)

4b). Using part (a), find all values of e and f so that this system has

(i) a unique solution, $\Leftrightarrow$ rank A = rank $[A|b] = 3$, so

$\Leftrightarrow \quad e \neq 4$.

(ii) infinitely many solutions, ~~or~~ $\Leftrightarrow$ rank A = rank $[A|b] < 3$, so

$\Leftrightarrow \quad e = 4$ and $f = -6$

(iii) no solutions. $\Leftrightarrow$ rank $A <$ rank $[A|b]$, so

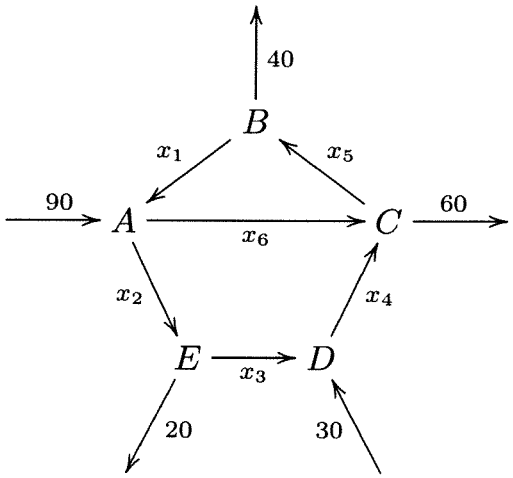
$\Leftrightarrow \quad e = 4$ and $f \neq -6$.

4c). In case b(ii) above, give a complete geometric description of the set of solutions.

If $e = 4, f = -6$, $[A|b] \sim \left[\begin{array}{ccc|c} 1 & 0 & 1 & -1 \\ 0 & 1 & -1 & 2 \\ 0 & 0 & 0 & 0 \end{array}\right]$; $\begin{array}{l} x = -1 - \lambda \\ y = 2 + \lambda \\ z = \lambda \end{array}$, $\lambda \in \mathbb{R}$.

Thus the set of solutions is the line in $\mathbb{R}^3$ through $(-1, 2, 0)$ with direction vector $(-1, 1, 1)$.

5. Consider the network of streets with intersections A, B, C, D and E below. The arrows indicate the direction of traffic flow along the **one-way streets**, and the numbers refer to the **exact** number of cars observed to enter or leave A, B, C, D and E during one minute. Each x_i denotes the unknown number of cars which passed along the indicated streets during the same period.



a) Write down a system of linear equations which describes the the traffic flow, together with all th constraints on the variables x_i, $i = 1, \ldots, 6$.

(Do not perform any operations on your equations: this is done for you in (b). Do not simply copy out the equations implicit in (b). You will not get any marks if you do this.)

Intersection	Flow in = Flow out
A	$90 + x_1 = x_2 + x_6$
B	$x_5 = x_1 + 40$
C	$x_4 + x_6 = x_5 + 60$
D	$30 + x_3 = x_4$
E	$x_2 = x_3 + 20.$

One-way streets : so $x_i \geq 0$, $i = 1, \ldots, 6$

of cars : so $x_i \in \mathbb{Z}$

(or : simply $x_i \in \mathbb{N}$)

b) The reduced row-echelon form of the augmented matrix of the system in part (a) is

$$\begin{array}{c} \\ \\ \\ \\ \\ \\ \end{array}\!\!\begin{array}{c} \begin{matrix} & & & & \Delta & t \end{matrix} \\ \left[\begin{array}{cccccc|c} 1 & 0 & 0 & 0 & -1 & 0 & -40 \\ 0 & 1 & 0 & 0 & -1 & 1 & 50 \\ 0 & 0 & 1 & 0 & -1 & 1 & 30 \\ 0 & 0 & 0 & 1 & -1 & 1 & 60 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right] \end{array}$$

Give the general solution. (Ignore the constraints from (a) at this point.)

$x_1 = -40 + \Delta$
$x_2 = 50 + \Delta - t$
$x_3 = 30 + \Delta - t$
$x_4 = 60 + \Delta - t$; $\Delta, t \in \mathbb{R}$
$x_5 = \Delta$
$x_6 = t$

c) If ED were closed due to roadwork, find the minimum flow along AC, **using your results from (b).**

(You must justify all your answers.)

ED closed $\Leftrightarrow x_3 = 0 \Leftrightarrow \Delta - t = -30$. Then

Constraints ($x_i \geq 0$)

$x_1 = -40 + \Delta \geq 0 \Leftrightarrow \Delta \geq 40$

$x_2 = 50 - 30 = 20$ ✓

$x_3 = 60 - 30 = 30$ ✓

$x_5 = \Delta \geq 0$

$x_6 = \Delta - 30 \geq 0 \Leftrightarrow \Delta \geq 30$

$\Rightarrow \Delta \geq 40$.

Flow along AC is $x_6 = \Delta + 30 \therefore x_6 \geq 70$

The minimum flow along x_6 is 70.

6. State whether each of the following statements is (always) true, or is (possibly) false, in the box after the statement.

- If you say the statement may be false, you must give an explicit example - with numbers, matrices, or functions, as is appropriate.
- If you say the statement is always true, you must give a clear explanation.

a) The kernel (nullspace) of the matrix $\begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ has dimension 1.

$A =$

$\dim \ker A = \#\text{cols} - \operatorname{rank} A = 3 - 2 = 1$

ANSWER TRUE

b) If a linear system is inconsistent, it cannot be homogeneous.

Every homogeneous linear system $AX=0$ has the solution $X=0$, and so is consistent.

ANSWER TRUE

6 (cont.).

c) If A is a 2 by 2 matrix and $A^2 = 0$, then $A = 0$.

eg If $A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$, $A^2 = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

but $A \neq 0$

ANSWER FALSE

d) If the coefficient matrix of a linear system of 2 equations in three variables has a row of zeros, the system has infinitely many solutions.

e.g. $\left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right]$ is inconsistent

ANSWER FALSE